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[redacted]
Medimmune UK
Aaron Klug Building
Granta Park
Cambridge
CB21 6GH
United Kingdom

Date Received:	10 May 2019
Date Tested:	22 May 2019
Date Test Completed:	24 Jun 2019
Purchase Order:	8300331868

CERTIFICATE OF ANALYSIS

Laboratory Reference Number: 0048626/1
Test Required: Preservative Efficacy Test
Date Received: 10/05/2019
Test Article: P18-028 MEDI0382 HC PET Study
Sample Code: Sample 1 - 1mg/mL M0382 (t=6M, 5C, 0.31% w/v)
Batch Ref: P18-028 (Stability at 5C, T=6M)
Qty Received: 1 x 55mL Blue-Capped USP Type 1 Glass Duran Bottle

Test	Method Item	Result
Pseudomonas aeruginosa	MM012/PS	Pass
Staphylococcus aureus	MM012/SA	Fail criteria A, pass criteria B for EP. Pass USP
Escherichia coli	MM012/EC	Pass
Aspergillus brasiliensis	MM012/AB	Pass
Candida albicans	MM012/CA	Pass
Pharmacopoeial Reference	MQ005/05	Ph Eur 9.0 5.1.3 & USP 42 <51>
Product Type	MQ005/10	Parenteral
Product Standard Data Sheet	FG047/psd	MDIM06-02

Comments
See attached table.

Approval is provided by Electronic Signature. Their name and position is shown below.

[redacted]

Date: 19 Jul 2019 09:13:01

Laboratary Manager - Pharmaceutical Microbiology

GMP Compliant Laboratory GLP Compliant Laboratory FDA Audited Laboratory

Company Registered in England No. 752951

Wickham Laboratories
Contract Analytical Services

PRESERVATIVE EFFICACY TEST

Sample Booking Number: 0048626

Test Article : P18-028 MEDI0382 HC PET Study
Sample Code : Sample 1 – 1mg/mL M0382 (t=6M, 5C, 0.31% w/v)
Batch Reference: : P18-028 (Stability at 5C, T=6M)
Laboratory Reference : 0048626/1

ORGANISM	PEPTONE/ SALINE CONTROL	COLONY FORMING UNITS PER ML OF SAMPLE					
		CONTACT TIME					
		0 HOURS	6 HOURS	24 HOURS	7 DAYS	14 DAYS	28 DAYS
Pseudomonas aeruginosa NCIMB 8626 **(EP & USP)**	1.9E5	1.5E5	<1.0E1	<1.0E1	<1.0E1	<1.0E1	<1.0E1
Staphylococcus aureus NCTC 10788 **(EP & USP)**	6.7E5	5.5E5	5.6E4	<1.0E1	<1.0E1	<1.0E1	<1.0E1
Escherichia coli NCIMB 8545 **(USP Only)**	2.8E5	3.1E5			5.5E1	8.0E1	<1.0E1
Aspergillus brasiliensis IMI 149007 **(EP & USP)**	1.5E5	1.4E5			<1.0E1	<1.0E1	<1.0E1
Candida albicans NCPF 3179 **(EP)**	3.7E5	3.5E5			<1.0E1	<1.0E1	<1.0E1
Candida albicans NCPF 3179 **(USP)**	3.7E5	3.5E5			<1.0E1	<1.0E1	<1.0E1

[redacted]
Documentation Manager – Microbiology

Date 19/07/19

Report No.: RM2861 QA Approved and date effective 08 April 2013

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Date Received:	10 May 2019
Date Tested:	22 May 2019
Date Test Completed:	24 Jun 2019
Purchase Order:	8300331868

CERTIFICATE OF ANALYSIS

Laboratory Reference Number: 0048626/2
Test Required: Preservative Efficacy Test
Date Received: 10/05/2019
Test Article: P18-028 MEDI0382 HC PET Study
Sample Code: Sample 2 - 5mg/mL M0382 (t=6M, 5C, 0.31% w/v)
Batch Ref: P18-028 (Stability at 5C, T=6M)
Qty Received: 1 x 55mL Blue-Capped USP Type 1 Glass Duran Bottle

Test	Method Item	Result
Pseudomonas aeruginosa	MM012/PS	Pass
Staphylococcus aureus	MM012/SA	Fail criteria A, pass criteria B for EP. Pass USP
Escherichia coli	MM012/EC	Pass
Aspergillus brasiliensis	MM012/AB	Pass
Candida albicans	MM012/CA	Pass
Pharmacopoeial Reference	MQ005/05	Ph Eur 9.0 5.1.3 & USP 42 <51>
Product Type	MQ005/10	Parenteral
Product Standard Data Sheet	FG047/psd	MDIM06-02

Comments
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Contract Analytical Services

PRESERVATIVE EFFICACY TEST

Sample Booking Number: 0048626

Test Article : P18-028 MEDI0382 HC PET Study
Sample Code : Sample 2 – 5mg/mL M0382 (t=6M, 5C, 0.31% w/v)
Batch Reference: : P18-028 (Stability at 5C, T=6M)
Laboratory Reference : 0048626/2

ORGANISM	PEPTONE/ SALINE CONTROL	COLONY FORMING UNITS PER ML OF SAMPLE					
		CONTACT TIME					
		0 HOURS	6 HOURS	24 HOURS	7 DAYS	14 DAYS	28 DAYS
Pseudomonas aeruginosa NCIMB 8626 **(EP & USP)**	1.9E5	3.5E5	<1.0E1	<1.0E1	<1.0E1	<1.0E1	<1.0E1
Staphylococcus aureus NCTC 10788 **(EP & USP)**	6.7E5	6.3E5	1.0E5	1.3E3	<1.0E1	<1.0E1	<1.0E1
Escherichia coli NCIMB 8545 **(USP Only)**	2.8E5	3.4E5			2.2E3	3.6E3	3.1E3
Aspergillus brasiliensis IMI 149007 **(EP & USP)**	1.5E5	1.3E5			<1.0E1	<1.0E1	<1.0E1
Candida albicans NCPF 3179 **(EP)**	3.7E5	4.4E5			<1.0E1	<1.0E1	<1.0E1
Candida albicans NCPF 3179 **(USP)**	3.7E5	4.4E5			<1.0E1	<1.0E1	<1.0E1

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Documentation Manager – Microbiology

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